

Klein-Gordon equation

Start with

$$E = \sqrt{p^2 c^2 + m^2 c^4}$$

Square both sides.

$$E^2 = p^2 c^2 + m^2 c^4$$

Substitute $i\hbar\partial/\partial t$ for E and $-i\hbar\nabla$ for p .

$$\left(i\hbar\frac{\partial}{\partial t}\right)^2 = (-i\hbar\nabla)^2 c^2 + m^2 c^4$$

By $i^2 = -1$

$$-\hbar^2 \frac{\partial^2}{\partial t^2} = -\hbar^2 c^2 \nabla^2 + m^2 c^4$$

Rearrange as

$$\hbar^2 \left(-\frac{\partial^2}{\partial t^2} + c^2 \frac{\partial^2}{\partial x^2} + c^2 \frac{\partial^2}{\partial y^2} + c^2 \frac{\partial^2}{\partial z^2} \right) = m^2 c^4$$

We seek ψ that solves the Klein-Gordon equation

$$\hbar^2 \left(-\frac{\partial^2}{\partial t^2} + c^2 \frac{\partial^2}{\partial x^2} + c^2 \frac{\partial^2}{\partial y^2} + c^2 \frac{\partial^2}{\partial z^2} \right) \psi = m^2 c^4 \psi$$

Let P be the momentum vector

$$P = \begin{pmatrix} E/c \\ p_x \\ p_y \\ p_z \end{pmatrix}$$

Let X be the position vector

$$X = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

Let ψ be the scalar function

$$\psi = A \exp\left(-\frac{i}{\hbar} P \cdot X\right) + B \exp\left(\frac{i}{\hbar} P \cdot X\right)$$

where

$$P \cdot X = P^\mu g_{\mu\nu} X^\nu = Et - p_x x - p_y y - p_z z$$

Show that ψ solves the Klein-Gordon equation.

Verify dimensions.

$$\hbar^2 \frac{\partial^2}{\partial t^2} = [\text{J s}]^2 \frac{1}{[\text{s}]^2} = [\text{J}]^2$$

$$\hbar^2 c^2 \frac{\partial}{\partial x^2} = [\text{J s}]^2 \frac{[\text{m}]^2}{[\text{s}]^2} \frac{1}{[\text{m}]^2} = [\text{J}]^2$$

$$m^2 c^4 = [\text{J}]^2$$

Eigenmath script